

Fundamentals of Linear Algebra for AI

Invertibility and Singular Value Decomposition (SVD)

DR. Mustafa M. Shiple

National Telecommunication Institute (NTI)

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2 (Pseudo inverse (A^+))



Invertibility



The Inverse Matrix ($\mathbf{A}^{-1}$)

- **The Goal:** The matrix inverse is the mathematical operation that **undoes** matrix multiplication.
- **Analogy:** It is the matrix equivalent of division in ordinary algebra (multiplying by $\mathbf{A}^{-1}$ is like dividing by $\mathbf{A}$).
- **Mathematical Definition:** If $\mathbf{A}$ is a square matrix, its inverse $\mathbf{A}^{-1}$ satisfies:

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

- If $\mathbf{A}^{-1}$ exists, the unique solution to $\mathbf{Ax} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

The Identity Matrix ($\mathbf{I}$)

The identity matrix is the multiplicative neutral element in matrix algebra, acting as "1" does in scalar algebra.

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. Conditions for the Inverse to Exist (Invertibility)

- Not all matrices have an inverse. A matrix that possesses an inverse is called **invertible** or **non-singular**.
- ① **The Dimension Condition: It Must Be Square**
 - ▶ The number of rows must equal the number of columns (e.g., 3×3).
- ② **The Determinant Condition: Its Determinant Must Be Non-Zero**
 - ▶ This is the most crucial condition: $\det(\mathbf{A}) \neq 0$.
 - ▶ **Geometric Interpretation:** If $\det(\mathbf{A}) = 0$, the matrix collapses the space (e.g., transforms 3D space onto a 2D plane). This kind of transformation **cannot be undone**.



Calculation Methods (2x2)



3. Calculating the Inverse: The 2×2 Case

The 2×2 matrix is the simplest case and has a direct formula:

- If the matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

- Then its inverse $\mathbf{A}^{-1}$ is given by the formula:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- **Recall:** The determinant $\det(\mathbf{A}) = ad - bc$.

Example

Let $\mathbf{A} = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$.

- $\det(\mathbf{A}) = (4)(6) - (7)(2) = 24 - 14 = 10$.
- $\mathbf{A}^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$.



Calculation Methods (Larger matrices)



4. Calculating the Inverse: The Gauss-Jordan Method

- This method is used for larger matrices (3×3 and up) and is generally the most computationally effective.
- **The Idea:** Apply Elementary Row Operations (EROs) to transform the matrix into the Identity Matrix.
- ① **Setup:** Write the matrix **A** next to the Identity Matrix **I** in an augmented matrix form:

$$[\mathbf{A}|\mathbf{I}]$$

- ② **Operations:** Apply EROs (scaling, addition, swapping rows) across the entire augmented matrix.
- ③ **Result:** Continue the operations until the left side is transformed into **I**. The resulting right side is the inverse $\mathbf{A}^{-1}$:

$$\xrightarrow{\text{Elementary Row Operations}} [\mathbf{I}|\mathbf{A}^{-1}]$$

1. Introduction to Gauss-Jordan Elimination

- **What it is:** A systematic algorithm to solve systems of linear equations or find the inverse of a matrix.
- **Core Idea:** Transform an augmented matrix into **Reduced Row Echelon Form (RREF)** using Elementary Row Operations (EROs).
- **Why RREF?** In RREF, the solution to the system (or the inverse matrix) can be read directly from the augmented matrix.

Key Concepts Involved

- **Augmented Matrix:** Combines the coefficient matrix with the constant vector (e.g., $[A|b]$).
- **Elementary Row Operations (EROs):** The allowed manipulations:
 - 1 Swapping two rows.
 - 2 Multiplying a row by a non-zero scalar.
 - 3 Adding a multiple of one row to another row.

2. General Steps of Gauss-Jordan Elimination

1 Form the Augmented Matrix:

- ▶ For solving $\mathbf{Ax} = \mathbf{b}$: Create $[\mathbf{A}|\mathbf{b}]$.
- ▶ For finding $\mathbf{A}^{-1}$: Create $[\mathbf{A}|\mathbf{I}]$, where $\mathbf{I}$ is the identity matrix.

2 Achieve Row Echelon Form (REF):

- ▶ Use EROs to get leading 1s (pivots) in each non-zero row.
- ▶ Ensure all entries below a pivot are zero.
- ▶ Ensure each leading 1 is to the right of the leading 1 in the row above it.

3 Achieve Reduced Row Echelon Form (RREF):

- ▶ After REF, use EROs to make all entries **above** each pivot also zero.
- ▶ (This is the "Jordan" step, as Gaussian Elimination only goes to REF).

4 Read the Result:

- ▶ If solving $\mathbf{Ax} = \mathbf{b}$, the right side of $[\mathbf{I}|\mathbf{x}]$ is the unique solution $\mathbf{x}$.
- ▶ If finding $\mathbf{A}^{-1}$, the right side of $[\mathbf{I}|\mathbf{A}^{-1}]$ is the inverse matrix.

Numerical example (1)

The System

We want to solve the following system:

$$x + 2y - z = 2 \quad (1)$$

$$3x + 8y + 2z = 11 \quad (2)$$

$$4x + 9y - z = 9 \quad (3)$$

- **Augmented Matrix** $[A|b]$:
$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 3 & 8 & 2 & 11 \\ 4 & 9 & -1 & 9 \end{array} \right)$$

- **Operation 1:** Eliminate the 3 and 4 in Column 1.

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 - 4R_1$$

Initial Setup and Forward Elimination

- **Result (First Pivot Set):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & \mathbf{2} & 5 & 5 \\ 0 & 1 & 3 & 1 \end{array} \right)$$

Completing Elimination (Row Echelon Form)

- **Operation 2:** Get a '1' in the pivot position (2, 2). Swap or scale. We'll swap for simplicity.

$$R_2 \leftrightarrow R_3$$

- **Result (Pivot 2 Set):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 2 & 5 & 5 \end{array} \right)$$

- **Operation 3:** Eliminate the 2 below the new pivot.

$$R_3 \leftarrow R_3 - 2R_2$$

- **Result (Row Echelon Form - REF):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & \mathbf{-1} & 3 \end{array} \right)$$

Backward Elimination (Reduced Row Echelon Form)

- **Operation 4:** Normalize the third pivot. $R_3 \leftarrow -1R_3$

- **Result (Pivots are all 1s):**
$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & \mathbf{1} & -3 \end{array} \right)$$

- **Operation 5:** Eliminate above the third pivot.

$$\begin{aligned} R_1 &\leftarrow R_1 + R_3 \\ R_2 &\leftarrow R_2 - 3R_3 \end{aligned}$$

- **Result (Final RREF):**
$$\left(\begin{array}{ccc|c} \mathbf{1} & 2 & 0 & -1 \\ 0 & \mathbf{1} & 0 & 10 \\ 0 & 0 & \mathbf{1} & -3 \end{array} \right)$$

Solution and Final RREF

Final Solution

The RREF matrix $[I|\mathbf{x}]$ gives the solution vector $\mathbf{x}$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -21 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & -3 \end{array} \right)$$

Thus, the unique solution is:

$$x = -21, \quad y = 10, \quad z = -3$$

Solution verification

Proof by Matrix-Vector Multiplication

To verify, we check if $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 2 & -1 \\ 3 & 8 & 2 \\ 4 & 9 & -1 \end{pmatrix} \begin{pmatrix} -21 \\ 10 \\ -3 \end{pmatrix} = \begin{pmatrix} (1)(-21) + (2)(10) + (-1)(-3) \\ (3)(-21) + (8)(10) + (2)(-3) \\ (4)(-21) + (9)(10) + (-1)(-3) \end{pmatrix}$$

Verification of the Solution

The Final Check

Continuing the matrix multiplication:

$$\mathbf{Ax} = \begin{pmatrix} -21 + 20 + 3 \\ -63 + 80 - 6 \\ -84 + 90 + 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 11 \\ 9 \end{pmatrix}$$

Conclusion

Since the result $\begin{pmatrix} 2 \\ 11 \\ 9 \end{pmatrix}$ matches the original right-hand side vector $\mathbf{b}$, the solution is verified and the ****Gauss-Jordan Elimination**** was performed correctly.

Numerical Example (2)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 2 & 2 \\ 4 & 5 \end{pmatrix}$$

Solution

- $\det(A) = (2 \times 5) - (2 \times 4) = 2 > 0$
- $$\begin{pmatrix} 2 & 2 & | & 1 & 0 \\ 4 & 5 & | & 0 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 2 & 2 & | & 1 & 0 \\ 0 & 1 & | & -2 & 1 \end{pmatrix} \xrightarrow{R_1 - 2R_2} \begin{pmatrix} 2 & 0 & | & 5 & -2 \\ 0 & 1 & | & -2 & 1 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & | & 2.5 & -1 \\ 0 & 1 & | & -2 & 1 \end{pmatrix}$$

verification

$$\mathbf{Ax} = \begin{pmatrix} 2 & 2 \\ 4 & 5 \end{pmatrix} \times \begin{pmatrix} 2.5 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Numerical Example (3)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Solution

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 - 4R_1 \\ R_3 - 7R_1}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & -6 & -12 & -7 & 0 & 1 \end{array} \right] \\ \xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right].$$

verification

Since we have row-reduced the matrix on the left so that it has a zero row, we conclude that it is not invertible (dependency).

Numerical Example (4)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$$

Solution

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{\substack{R_2 - R_1 \\ R_3 - R_1}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array} \right] &\xrightarrow{\substack{R_1 - R_2 \\ R_3 - 2R_2}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array} \right] \\ &\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array} \right] &\xrightarrow{\substack{R_1 + 2R_3 \\ R_2 - 3R_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5/2 & 4 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array} \right] \end{aligned}$$

Numerical Example (5)

The System and Initial Augmented Matrix

Consider the system:

$$x - 3y = -5$$

$$-2x + 7y = 12$$

→ Initial Matrix: $\left(\begin{array}{cc|c} 1 & -3 & -5 \\ -2 & 7 & 12 \end{array} \right)$

❶ **Step 1: Eliminate below the first pivot.**

$$R_2 \leftarrow R_2 + 2R_1 \quad (\text{Get 0 below the first } 1)$$

$$\left(\begin{array}{cc|c} 1 & -3 & -5 \\ 0 & 1 & 2 \end{array} \right) \quad (\text{Now in Row Echelon Form - REF})$$

❷ **Step 2: Eliminate above the second pivot (Jordan step).**

$$R_1 \leftarrow R_1 + 3R_2 \quad (\text{Get 0 above the second } 1)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right) \quad (\text{Now in Reduced Row Echelon Form - RREF})$$

Solving a System of Equations

Final Solution

The RREF matrix immediately gives the solution:

$$x = 1$$

$$y = 2$$



Application and Solution



Application: Solving Systems of Linear Equations

- The matrix inverse is the most powerful tool for solving a system of linear equations $\mathbf{Ax} = \mathbf{b}$ when a unique solution exists.

- **The Steps:**

- 1 Start with the system: $\mathbf{Ax} = \mathbf{b}$
- 2 Multiply both sides of the equation on the left by $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1}(\mathbf{Ax}) = \mathbf{A}^{-1}\mathbf{b}$$

- 3 Since $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$:

$$\mathbf{Ix} = \mathbf{A}^{-1}\mathbf{b}$$

- **The Final Solution is:**

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

28. When No Exact Solution Exists

- In Machine Learning, $\mathbf{Ax} = \mathbf{b}$ rarely has an exact solution due to noise/errors.
- We seek the ****best approximate solution****.
- This is where the **Projection** concept comes in.

29. Least Squares Approximation

- Goal: Find $\mathbf{x}^*$ that minimizes the error $\|\mathbf{Ax} - \mathbf{b}\|^2$.
- The solution is given by the **Normal Equation**:

$$\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$$

- **AI Context:** The mathematical foundation for linear regression (finding the "line of best fit").



(Pseudo inverse (A^+))





(Non-singular inverse)



1. Definition and Motivation

When the Standard Inverse Fails

- The *Pseudoinverse $(A^+)^*$, or Moore-Penrose Inverse, is a generalization of the standard inverse (A^{-1}) .
- It exists for **any matrix** A :
 - ▶ Rectangular matrices ($m \neq n$).
 - ▶ Square, but singular (rank-deficient) matrices.
- **Goal:** To find the "best approximate solution," $\mathbf{x}^*$, for the system $A\mathbf{x} \approx \mathbf{b}$.

2. Primary Applications of A^+

- **Overdetermined Systems ($m > n$):**

- ▶ More equations than unknowns (e.g., linear regression).
- ▶ No exact solution usually exists.
- ▶ A^+ finds the vector $\mathbf{x}^*$ that minimizes the error $\|A\mathbf{x} - \mathbf{b}\|^2$.
- ▶ This is the **Least Squares Solution**.

- **Underdetermined Systems ($m < n$):**

- ▶ Fewer equations than unknowns.
- ▶ Infinitely many solutions exist.
- ▶ A^+ finds the solution $\mathbf{x}^*$ with the minimum length ($\|\mathbf{x}^*\|^2$).
- ▶ This is the **Minimum Norm Solution**.

Calculation: Full Rank Cases (One-Sided Inverses)

Simple Formulas when the Rank is Complete

The solution is always $\mathbf{x}^* = \mathbf{A}^+\mathbf{b}$.

Case	Condition	Formula for $\mathbf{A}^+$
Full Column Rank	$\text{rank}(A) = n \ (m > n)$	$(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$
Full Row Rank	$\text{rank}(A) = m \ (m < n)$	$\mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}$
Non-Singular	$\text{rank}(A) = m = n$	$\mathbf{A}^{-1}$

Note: These simplified formulas only work if A has full rank.

Full Column Rank Example

Calculating $\mathbf{A}^+$ for Least Squares

We use the formula: $\mathbf{A}^+ = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ for the 3×2 matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (\text{Full Column Rank, rank} = 2)$$

① Calculate $\mathbf{A}^T \mathbf{A}$:

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

② Calculate $(\mathbf{A}^T \mathbf{A})^{-1}$: ($\det = 3$)

$$(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

③ Calculate $\mathbf{A}^+$:

$$\mathbf{A}^+ = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 1/3 & -1/3 \\ -1/3 & 1/3 & 2/3 \end{pmatrix}$$

Ranked Matrices

For an $m \times n$ matrix A (m rows, n columns), the maximum possible rank is $\min(m, n)$. If a matrix is rank-deficient (i.e., its rank is less than $\min(m, n)$), it means its columns are linearly dependent and its rows are linearly dependent. In this case:

- Both $\mathbf{A}^T \mathbf{A}$ and $\mathbf{A} \mathbf{A}^T$ will be singular matrices.
- The inverses $(\mathbf{A}^T \mathbf{A})^{-1}$ and $(\mathbf{A} \mathbf{A}^T)^{-1}$ will not exist.



The SVD Method



Prerequisites for SVD

- An intuitive understanding of:
 - ▶ **Diagonal Matrix** (stretches axes).
 - ▶ **Orthogonal Matrix** (produces rotation).
- The main appeal of SVD is that it generalizes to **rectangular matrices**.
- A vector in R^2 (e.g., $(1, 2)$) is fundamentally different from a vector in R^3 (e.g., $(1, 2, 0)$).
- The R^2 vector is **always devoid** of the third dimension.
- A rectangular matrix can transform a vector from one dimension to another.
- Example: A 2×3 matrix transforms a vector in R^3 **down** to a vector in R^2 .

Visualizing the Simplest Case

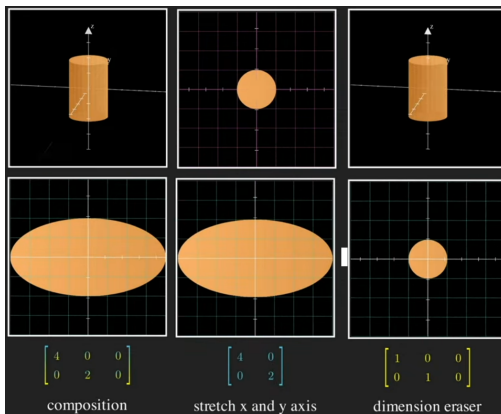
- Consider the "rectangular identity matrix" or **Dimension Eraser**.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

- This matrix transforms $R^3 \rightarrow R^2$.
- It **preserves X and Y** but completely **removes the Z value**.
- Vector $(1, 2, 1)$ in R^3 is mapped down to $(1, 2)$ in R^2 .
- ALL vectors of the form $(1, 2, \mathbf{Z})$ transform to $(1, 2)$.

Composition with Rectangular Matrices

- Multiplying the Dimension Eraser with a Diagonal Matrix:
- $C = D \times \text{Eraser}$
- C first removes the 3rd dimension, then stretches the X and Y axes. C encapsulates both.





Constructing Symmetry



Symmetric Matrix Flashback

- A **Symmetric Matrix** is a square matrix where entries across the diagonal are identical ($A = A^T$).
- Its eigenvectors are guaranteed to be **perpendicular** (orthogonal).
- Normalizing and packaging the perpendicular eigenvectors creates an **Orthogonal Matrix**.
- An orthogonal matrix implies **rotation**.

Naming the Symmetric Products

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$A_{2 \times 3} \qquad A^T_{3 \times 2} \qquad AA^T_{2 \times 2}$

- For a 2×3 matrix A :
- $S_{\text{left}} = AA^T$ (a 2×2 symmetric matrix).
- S_{left} has **two perpendicular eigenvectors** (in R^2).
- Eigenvectors of S_{left} are the **Left Singular Vectors** of A .

Naming the Symmetric Products

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

$A^T \quad 3 \times 2 \qquad A \quad 2 \times 3 \qquad A^T A \quad 3 \times 3$

- For a 2×3 matrix A :
- $S_{\text{right}} = A^T A$ (a 3×3 symmetric matrix).
- S_{right} has **three perpendicular eigenvectors** (in R^3).
- Eigenvectors of S_{right} are the **Right Singular Vectors** of A .

The General SVD Method

The Most Robust Method for all Matrices

- ① **Decompose** A : Use Singular Value Decomposition (SVD):

$$A = U\Sigma V^T$$

- ② **Calculate** Σ^+ : Form the pseudoinverse of the diagonal matrix Σ :

- ▶ Transpose Σ .
- ▶ Replace every non-zero singular value σ_i with its reciprocal, $1/\sigma_i$.

- ③ **Assemble** A^+ : The final result is:

$$A^+ = V\Sigma^+U^T$$

The Four Moore-Penrose Conditions

Defining Properties of A^+

The unique matrix A^+ must satisfy the following four properties:

① **Inverse Property:**

$$AA^+A = A$$

② **Pseudoinverse Property:**

$$A^+AA^+ = A^+$$

③ **Symmetry I:**

$$(AA^+)^T = AA^+$$

④ **Symmetry II:**

$$(A^+A)^T = A^+A$$

These conditions guarantee the uniqueness of the pseudoinverse.

Numerical Example: SVD

Decomposing a Simple 2×2 Matrix

Let's find the SVD of the rank-1 matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

① Find Σ and V from $\mathbf{A}^T \mathbf{A}$:

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 0$.

► Singular Values: $\sigma_1 = \sqrt{2}$, $\sigma_2 = 0$.

► Right Singular Vectors (V): $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

② Find U from $\mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i$:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

(The remaining vector $\mathbf{u}_2$ is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, completing the basis.)

Numerical Example: SVD

Decomposing a Simple 2×2 Matrix

Let's find the SVD of the rank-1 matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

- ① Find Σ and V from $A^T A$:

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 0$.

- ② Find U from $\mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i$:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

(The remaining vector $\mathbf{u}_2$ is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, completing the basis.)

Final Decomposition $\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^T$:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

Singular Value Decomposition (SVD)

- The most general form of matrix decomposition, applicable to *any* matrix $\mathbf{A}$ ($m \times n$).

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

- $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices.
- $\mathbf{\Sigma}$ is a diagonal matrix containing the **Singular Values** (σ_i).

The Three Components: $A = U\Sigma V^T$

U

- $m \times m$ matrix
- **Orthogonal**
($U^T U = I$)
- Columns are the "Left Singular Vectors"
- Forms an orthonormal basis for the output space ($\mathbb{R}^m$)

Σ (Sigma)

- $m \times n$ matrix
- **Diagonal**
- Diagonal entries σ_i are the "Singular Values"
- $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$

V^T (V-Transpose)

- V is an $n \times n$ matrix
- **Orthogonal**
($V^T V = I$)
- Columns of V are the "Right Singular Vectors"
- V^T contains an orthonormal basis for the input space ($\mathbb{R}^n$)

Why is SVD so Important?

Absolute Generality

Unlike other factorizations (like eigendecomposition), SVD works on **any matrix** A .

- Square or rectangular
- Full rank or rank-deficient
- Invertible or singular

Applications

SVD is the mathematical engine behind:

- Principal Component Analysis (PCA)
- Image and data compression (Low-Rank Approximation)
- Recommender systems (Netflix Prize)
- Natural Language Processing (LSA)

The Solution: A 3-Step Plan to find U, Σ, V

- 1 **Find V and Σ from $A^T A$.** Compute the $n \times n$ symmetric matrix $A^T A$. Its eigenvalues and eigenvectors give us V and Σ .
- 2 **Find U from AA^T (or a shortcut).** Compute the $m \times m$ symmetric matrix AA^T . Its eigenvectors give us U .
- 3 **Assemble the pieces.** Carefully arrange the components, ensuring singular values are in descending order and eigenvectors match.

The Key Insight

$A^T A$ and AA^T are symmetric, so they have real eigenvalues and orthogonal eigenvectors (Spectral Theorem). SVD is built on this.

Application: Low-Rank Approximation

SVD writes A as a sum of "rank-1" matrices, weighted by their singular values:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T$$

Data Compression

The first singular value σ_1 is the largest and corresponds to the "most important" part of the matrix.

We can **approximate** A by keeping only the first k terms:

$$A \approx A_k = \sigma_1 u_1 v_1^T + \cdots + \sigma_k u_k v_k^T$$

This is the **best possible** rank- k approximation of A . This is how image, audio, and video compression works.

A Note on Rectangular Matrices

The process is identical, but the Σ matrix is rectangular ($m \times n$).

Example: A is 2×3

$A^T A$ is 3×3 (gives Σ with 3 σ_i 's, and V is 3×3). AA^T is 2×2 (gives U is 2×2). The Σ matrix will be 2×3 :

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{pmatrix}$$

The SVD still works perfectly. The SVD for A^T is $(V\Sigma^T U^T)$.

Conclusion

What is SVD?

The decomposition of any matrix A into $U\Sigma V^T$.

How to Solve it?

1. Find eigenvalues (λ_i) and eigenvectors (v_i) of $A^T A$. 2. Set $\sigma_i = \sqrt{\lambda_i}$ to build Σ . 3. Set v_i as columns to build V . 4. Find $u_i = \frac{1}{\sigma_i} A v_i$ to build U .

Why?

It reveals the fundamental geometry (rotation-scaling-rotation) and is the core of modern data science applications.

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4 Vector Spaces & Basis

5 Eigendecomposition & Applications



Vector Spaces & Basis



32. Vector Spaces

- A set of vectors that is **closed** under vector addition and scalar multiplication.
- If $\mathbf{u}$ and $\mathbf{v}$ are in the space, then $\mathbf{u} + \mathbf{v}$ and $c\mathbf{v}$ must also be in the space.
- The most common is $\mathbb{R}^n$, but subspaces are crucial.

33. Span and Linear Combinations

- A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is $\mathbf{y} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$.
- The **span** of a set of vectors is the set of all possible linear combinations of those vectors.
- It describes the full space (or subspace) that can be reached by those vectors.

34. Linear Dependence and Independence

- **Dependent:** A set of vectors is linearly dependent if one vector can be written as a linear combination of the others (i.e., they are redundant).
- **Independent:** No vector in the set can be written as a linear combination of the others. The vectors are essential.
- **AI Context:** Linear dependence suggests **redundant features** in a dataset.

35. Basis of a Vector Space

- A set of vectors is a **basis** if:
 - ① They are **linearly independent**.
 - ② They **span** the entire vector space.
- Every vector in the space can be written uniquely as a linear combination of the basis vectors.

36. The Four Fundamental Subspaces

- For any matrix $\mathbf{A}$, there are four key subspaces:
 - ① **Column Space** ($C(\mathbf{A})$)
 - ② **Null Space** ($N(\mathbf{A})$)
 - ③ **Row Space** ($C(\mathbf{A}^T)$)
 - ④ **Left Null Space** ($N(\mathbf{A}^T)$)

37. Column Space ($C(\mathbf{A})$)

- The span of the columns of $\mathbf{A}$.
- Represents the space of all possible output vectors $\mathbf{b}$ for $\mathbf{Ax} = \mathbf{b}$.
- Dimension of $C(\mathbf{A})$ is the rank of $\mathbf{A}$.

38. Null Space ($N(\mathbf{A})$)

- The set of all vectors $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{0}$.
- These vectors are "annihilated" by the transformation $\mathbf{A}$.
- Dimension of $N(\mathbf{A})$ is the nullity of $\mathbf{A}$.

39. The Rank-Nullity Theorem

- For an $m \times n$ matrix $\mathbf{A}$:

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n$$

- This fundamental theorem relates the dimensions of the column space and the null space.

40. Projection and Orthogonality (Revisited)

- The Column Space and the Left Null Space are **orthogonal complements**.
- The Row Space and the Null Space are **orthogonal complements**.
- This orthogonal decomposition is mathematically elegant and simplifies the Least Squares problem.

Table of Contents (V/V)

5 Eigendecomposition & Applications



Eigendecomposition & Applications



42. Eigenvalues and Eigenvectors

- For a square matrix $\mathbf{A}$, an **eigenvector** $\mathbf{v}$ is a non-zero vector such that when multiplied by $\mathbf{A}$, it only changes magnitude, not direction.

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- λ is the corresponding **eigenvalue** (the scalar magnitude change).

43. Interpretation in AI

- Eigenvectors represent the **principal directions** or axes of a linear transformation.
- Eigenvalues represent the **variance** or importance along those axes.
- High eigenvalues indicate directions where the data is most spread out.

44. Computing Eigendecomposition

- Finding λ : Solve the characteristic equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.
- Finding $\mathbf{v}$: Solve the linear system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$.

45. Diagonalization

- If $\mathbf{A}$ has a full set of n linearly independent eigenvectors, it is diagonalizable:

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

- $\mathbf{D}$ is a diagonal matrix of eigenvalues, and $\mathbf{P}$ is a matrix whose columns are the corresponding eigenvectors.
- Diagonalization simplifies calculating powers of a matrix ($\mathbf{A}^k$).

46. Principal Component Analysis (PCA)

- **Purpose:** Dimensionality reduction and feature extraction.
- **Process:**
 - 1 Compute the **Covariance Matrix** (**C**) of the data.
 - 2 Compute the **Eigenvectors** and **Eigenvalues** of **C**.
 - 3 The eigenvectors are the **Principal Components** (new, uncorrelated axes).
 - 4 The largest eigenvalues correspond to the most important axes.

49. Linear Algebra in Modern AI (LLMs)

- **Word Embeddings:** Words are converted into high-dimensional vectors.
- **Attention Mechanism (Transformers):** The core operation uses massive matrix multiplications to calculate relevance scores between all words in a sequence.
- **Storage and Computation:** Linear algebra is the language of modern deep learning hardware (TPUs, GPUs).

Thank You

- Linear Algebra is not just theory; it is the fundamental tool for handling, transforming, and interpreting data in AI.
- **Key Takeaways:** Vectors as features, Matrices as transformations, Eigenvectors/SVD for data reduction (PCA).
- **Next Steps:** Explore optimization (gradient descent) and advanced decomposition techniques.

Questions and Discussion